

Initial Plan

PROJECT NEXUS – CYBERSECURITY TRACK

PLATFORM: G0G NEXUS

TRACK: CYBERSECURITY (RED TEAM & BLUE TEAM)

FORMAT: GUIDED LABS + FLAG SUBMISSION (TRYHACKME-STYLE)

> Overview

The Cybersecurity Track of Project Nexus is designed as a hands-on, room-based learning experience inspired by platforms such as TryHackMe and Hack The Box. Each room focuses on a specific defensive or offensive security skill and uses open-source, safe-to-practice environments.

Participants will:

- Learn core cybersecurity concepts through guided instructions
- Apply skills using real tools and simulated vulnerable targets
- Submit flags or proof of completion via the Nexus Platform
- Earn Sparky Points (XP) upon successful completion

RED TEAM ROOMS

Red-1: Web Exploitation – SQL Injection Basics

Room Information

- Room Type: Red Team (Offensive Security)
- Difficulty: Beginner
- Estimated Time: 45–60 minutes
- Prerequisites: Basic web concepts (HTTP, forms, login pages)

External Learning Resource

Primary Lab Environment:

OWASP Juice Shop (Official Project)

- Documentation: <https://owasp.org/www-project-juice-shop/>
- Purpose: Intentionally vulnerable web application for learning OWASP Top 10

Learning Objectives

By the end of this room, participants should be able to:

- Understand what SQL Injection (SQLi) is and why it occurs
- Identify vulnerable input fields in web applications
- Perform a basic authentication bypass using SQL Injection
- Explain the real-world impact of SQL Injection vulnerabilities

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Lab Instructions

1. Open the provided OWASP Juice Shop instance.
2. Navigate to the Login page of the application.
3. Inspect the login form fields and observe how data is sent.
4. Attempt a basic SQL Injection payload in the email or username field.
5. Observe application responses and error messages.
6. Successfully bypass authentication and access a privileged account.

Milestone Check

To complete this room, the participant must:

- Successfully bypass the login mechanism using SQL Injection
- Gain access to admin-level page

Flag Submission

- Flag Format: **NEXUS{SQLI_SUCCESS}**
- Submission Method: Nexus Platform → Room Flag Input

Upon correct submission:

- Room is marked as Completed
- Confirmation message is shown
- +100 Sparky XP

BLUE TEAM ROOMS

Blue-1: Host & Network Hardening Fundamentals

Room Information

- Room Type: Blue Team (Defensive Security)
- Difficulty: Beginner to Intermediate
- Estimated Time: 60 minutes
- Prerequisites: Basic Linux or OS familiarity

External Learning Resource

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Primary Lab Environment:

- Local or Demo Linux Virtual Machine
- CIS Benchmarks (Reference)

Learning Objectives

By the end of this room, participants should be able to:

- Understand what host hardening means
- Identify insecure default system configurations
- Apply basic host and network security controls
- Document and justify security changes

Lab Instructions

1. Access the provided Linux VM or demo environment.
2. Review running services and open ports.
3. Disable or remove unnecessary services.
4. Configure basic firewall rules (allow/deny rules).
5. Apply secure account and password policies.
6. Verify changes and ensure system stability.

Milestone Check

Participant must demonstrate:

- Firewall enabled and active
- At least three (3) unnecessary services disabled
- Secure password or account policy applied

Flag Submission Options

Participants may submit one of the following:

- Flag: `NEXUS{HOST_HARDENED}`
- OR a short configuration summary (text-based proof)

Upon correct submission:

- Room is marked as Completed
- Confirmation message is shown
- +100 Sparky XP

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BLUE

PROJECT NEXUS – CYBERSECURITY TRACK

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Session 6:

Blue-1 — Host & Network Hardening

Baseline hardening, firewall rules, patch management, secure account policies.

- Implement baseline hardening on lab VM and document changes.

Session 8:

Blue-2 — Monitoring & Packet Analysis

Deep packet analysis to identify common attack patterns and IOCs.

- Capture traffic during a demo attack and detect indicators (SYN flood, suspicious HTTP payloads).

<https://blueteamlabs.online/home/challenge/network-analysis-ransomware-3dd520c7ec>

<https://blueteamlabs.online/home/challenge/network-analysis-malware-compromise-e882f32908>

<https://blueteamlabs.online/home/challenge/network-analysis-web-shell-d4d3a2821b>

Session 10:

Blue-3 — IDS/IPS & SIEM Fundamentals

Signature basics (Snort/Suricata), log ingestion, SIEM correlation rules.

- Deploy a simple detection signature and ingest sample logs into a lightweight SIEM (Wazuh/Security Onion demo).

Session 12:

Blue-4 — Incident Response & Cloud Defense

IR lifecycle, evidence preservation, cloud hardening (GCP IAM/buckets), alerting.

- Execute a short playbook on a simulated compromise; implement a secure bucket configuration

RED-1 FULL

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RED-1: Web Exploitation Lab Room

Room Information

Room Type: Red Team (Offensive Security)

Session Number: Session 5

Difficulty: Beginner

Estimated Time: 90-120 minutes

Prerequisites:

- Basic understanding of HTTP requests/responses
- Familiarity with web browsers and developer tools
- Basic knowledge of databases and SQL (helpful but not required)

Sparky Points: 100 XP

Learning Objectives

By the end of this room, participants will be able to:

1. Identify SQL injection vulnerabilities in web application login forms and search fields
2. Exploit SQL injection to bypass authentication mechanisms
3. Extract sensitive data from databases using UNION-based SQL injection
4. Understand the difference between Reflected and Stored XSS attacks
5. Execute XSS payloads to steal session cookies
6. Explain the real-world impact and remediation for OWASP Top 10 vulnerabilities

External Learning Resource: OWASP Juice Shop

What is OWASP Juice Shop?

Official Project: <https://owasp.org/www-project-juice-shop/>

OWASP Juice Shop is the ONLY external application you'll use for this entire room. It's:

- A modern, intentionally vulnerable web application
- Maintained by OWASP (Open Web Application Security Project)
- Written in Node.js, Express, and Angular
- Contains 100+ vulnerabilities from the OWASP Top 10
- Specifically designed for security training

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> Lab Environment Setup

MANDATORY: Use Hosted OWASP Juice Shop

You MUST use the official hosted instance for this lab.

Access URL: <https://juice-shop.herokuapp.com>

Alternative Official Instance: <https://demo.owasp-juice.shop>

> Tools Required

Required Tools

1. Web Browser

- Chrome or Firefox (required)
- Developer Tools (F12) enabled

2. Text Editor

- For copying payloads
- Notepad++, VS Code, or any text editor

Optional Tools (For Advanced Students)

Burp Suite Community Edition (not required, but helpful)

- Download: <https://portswigger.net/burp/communitydownload>
- Used to intercept HTTP requests
- We'll provide all payloads, so this is optional

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Flag System Explanation

How Flags Work in This Room

IMPORTANT: OWASP Juice Shop has its own built-in scoring system. Here's how we're using it:

Juice Shop Challenges → Nexus Flags

Each challenge you complete in Juice Shop unlocks a flag that you submit to Nexus.

Here's the process:

1. Complete challenge in Juice Shop
 - You'll see a green "Solved" notification
 - The Score Board updates automatically
2. Find your challenge code
 - Go to Account → Score Board
 - Find the completed challenge
 - Note the challenge name
3. Generate your flag
 - Use this formula: `NEXUS{JUICE_<CHALLENGE_NAME>_<YOUR_TEAM_ID>}`
 - We'll provide the exact flag format for each challenge below
4. Submit to Nexus platform
 - Go to Nexus Cybersec website
 - Navigate to RED-1 room
 - Enter the flag exactly as specified

Example:

Challenge Completed: "Login Admin"

Your Team ID: TEAM05

Flag Format: `NEXUS{JUICE_LOGIN_ADMIN_TEAM05}`

You submit: `NEXUS{JUICE_LOGIN_ADMIN_TEAM05}`

PROJECT NEXUS – CYBERSECURITY TRACK

PLATFORM: G06 NEXUS

TRACK: CYBERSECURITY (RED TEAM & BLUE TEAM)

Challenge 1: SQL Injection - Admin Login

Objective

Use SQL injection to log in as the administrator without knowing the password.

Background

SQL injection occurs when user input is directly concatenated into SQL queries. The vulnerable query looks like:

```
SELECT * FROM Users WHERE email = '[USER_INPUT]' AND password = '[PASSWORD_HASH]'
```

Instructions

Step 1: Navigate to Login Page

1. On Juice Shop homepage, click "Account" → "Login"
2. URL should be: <https://juice-shop.herokuapp.com/#/login>
3. Open Developer Tools (F12)
4. Go to Network tab

Step 2: Test for SQL Injection

Normal Login Attempt:

- Email: test@test.com
- Password: [test123](#)
- Click "Log in"
- Observe: "Invalid email or password" error

Step 3: SQL Injection Exploitation

The Payload:

In the Email field, enter:

[admin@juice-sh.op'--](#)

In the Password field, enter:

[anything](#)

Click Log in

What happens:

- The SQL query becomes:

```
SELECT * FROM Users WHERE email = 'admin@juice-sh.op'--' AND password = '...'
```
- The `--` comments out the rest of the query

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- Password check is bypassed!
- You're logged in as admin

Step 4: Verify Success

After successful login:

1. Check the top-right corner - you should see "admin@juice-sh.op"
2. Go to Account → Score Board
3. Find the challenge: "Login Admin" - it should now show  Solved

Step 5: Capture the Flag

Challenge Name in Juice Shop: Login Admin

Flag Format: NEXUS{SQLI_ADMIN_<YOUR_TEAM_ID>}

Example: If your team ID is TEAM05, submit:

NEXUS{SQLI_ADMIN_TEAM05}

Where to get YOUR_TEAM_ID:

- Check your Nexus profile
- Or ask your team lead
- Format: TEAM## (e.g., TEAM01, TEAM02, etc.)

Verification Checklist

- Successfully bypassed login without knowing password
- Logged in as admin@juice-sh.op
- "Login Admin" challenge shows as Solved on Score Board
- Submitted flag to Nexus platform

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Challenge 2: SQL Injection - Data Exfiltration

Objective

Use UNION-based SQL injection to extract hidden data from the database.

Background

UNION-based SQL injection allows combining results from multiple SELECT statements, enabling data extraction from other database tables.

Instructions

Step 1: Find the Search Function

1. On Juice Shop homepage, locate the search bar (top right)
2. Try a normal search: `apple`
3. Observe the URL: `https://juice-shop.herokuapp.com/#/search?q=apple`

Step 2: Test for SQL Injection

Normal Search:

`apple`

SQL Injection Test:

`apple' OR 1=1--`

If vulnerable, you'll see all products instead of just apple-related ones.

Step 3: Determine Column Count

Try different column counts with NULL values:

Test 1:

`' UNION SELECT NULL--`

If error: not enough columns

Test 2:

`' UNION SELECT NULL, NULL--`

If error: not enough columns

Keep adding NULLs until no error. For Juice Shop, you need 9 columns:

`' UNION SELECT NULL,NULL,NULL,NULL,NULL,NULL,NULL,NULL,NULL FROM Users--`

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Step 4: Extract User Emails

Now that you know the column count, extract data:

Payload:

```
' UNION SELECT NULL, email, password, NULL, NULL, NULL, NULL, NULL, NULL FROM Users--
```

Paste this in the search bar and press Enter.

What you'll see:

- Products displayed will include "fake products" with user emails in the name field
- You'll see emails like: `admin@juice-sh.op`, `jim@juice-sh.op`, etc.
- Password hashes will also be visible

Step 5: Count the Extracted Users

Count how many unique user emails you can see in the results.

Typical count: 15-20 users

Step 6: Verify Challenge Completion

1. Go to Account → Score Board
2. Look for challenge: "Database Schema" or "Login Jim" (related to user extraction)
3. The challenge should show  Solved

Step 7: Capture the Flag

Challenge Name: Varies, but related to user extraction

Flag Format: `NEXUS{SQLI_UNION_<NUMBER_OF_USERS>_<YOUR_TEAM_ID>}`

Example: If you found 17 users and your team is TEAM05:

```
NEXUS{SQLI_UNION_17_TEAM05}
```

Verification Checklist

- Successfully identified column count (9 columns)
- Extracted user emails using UNION SELECT
- Counted total number of users
- Submitted flag with correct user count

PROJECT NEXUS – CYBERSECURITY TRACK

PLATFORM: G0G NEXUS

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Challenge 3: Reflected XSS

Objective

Execute a reflected XSS attack by injecting JavaScript into the search parameter.

Background

Reflected XSS occurs when user input is immediately returned by the web application without sanitization, allowing script execution in the victim's browser.

Instructions

Step 1: Identify XSS Vulnerability

The search function is vulnerable to XSS.

Step 2: Test Basic XSS

In the search bar, enter:

```
<script>alert('XSS')</script>
```

What happens:

- An alert box pops up with "XSS"
- This confirms the vulnerability!

Step 3: Session Cookie Theft PoC

Let's steal the session cookie:

```
<script>alert(document.cookie)</script>
```

Result:

- Alert box shows your session token
- In real attacks, this would be sent to attacker's server

Step 4: Advanced XSS Payload

Try this payload:

```
<iframe src='javascript:alert("XSS")'>
```


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Step 5: Verify Challenge Completion

1. Go to Account → Score Board
2. Find challenge: "DOM XSS" or "Reflected XSS"
3. Should show  Solved

Step 6: Capture the Flag

Challenge Name in Juice Shop: DOM XSS or Bonus Payload

Flag Format: NEXUS{XSS_REFLECTED_<YOUR_TEAM_ID>}

Example:

NEXUS{XSS_REFLECTED_TEAM05}

Verification Checklist

- Successfully executed JavaScript in browser via search
- Displayed session cookie in alert box
- XSS challenge marked as Solved
- Submitted flag to Nexus

Challenge 4: Persistent XSS

Objective

Inject a persistent XSS payload that executes every time a page is loaded.

Background

Persistent (Stored) XSS is stored on the server and executes whenever the affected page is viewed. More dangerous than reflected XSS.

Instructions

Step 1: Navigate to Customer Feedback

1. Scroll to bottom of homepage
2. Click "Customer Feedback" link
3. Or go directly to: <https://juice-shop.herokuapp.com/#!/contact>

Step 2: Craft XSS Payload

In the Comment field, enter:

```
<iframe src="javascript:alert('Stored_XSS')">
```

Fill out other required fields:

- Rating: Select any rating

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- CAPTCHA: Solve the simple math problem

Click Submit

Step 3: Verify Persistence

1. Navigate away from the feedback page
2. Go back to Customer Feedback or refresh the page
3. The alert should NOT pop up again (Juice Shop sanitizes feedback display)

Alternative Method - Admin Panel XSS:

If you're logged in as admin:

1. Go to Administration → Customer Feedback
2. View all submitted feedback
3. Your XSS payload might execute in the admin panel!

Step 4: Verify Challenge Completion

1. Go to Account → Score Board
2. Find challenge: "API-only XSS" or "Client-side XSS Protection"
3. Check if marked  Solved

Step 5: Alternative Challenge - User Registration XSS

Try XSS in the registration form:

1. Go to Account → Login → Not yet a customer?
2. In the Email field, try:

```
<script>alert('XSS')</script>@juice-sh.op
```

Step 6: Capture the Flag

Challenge Name: API-only XSS or similar

Flag Format: NEXUS{XSS_STORED_<YOUR_TEAM_ID>}

Example:

```
NEXUS{XSS_STORED_TEAM05}
```

Verification Checklist

- Submitted XSS payload in feedback form
- Tested persistence (payload stored in database)
- Related XSS challenge marked as Solved
- Submitted flag to Nexus

PROJECT NEXUS – CYBERSECURITY TRACK

PLATFORM: G0G NEXUS

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Challenge 5: Authentication Bypass via Password Reset

Objective

Find and exploit a weakness in the password reset mechanism.

Background

Not all authentication bypasses involve SQL injection. Logic flaws in password reset, account creation, or session management can also lead to unauthorized access.

Instructions

Step 1: Initiate Password Reset

1. Go to login page: <https://juice-shop.herokuapp.com/#/login>
2. Click "Forgot your password?"
3. Enter an email: jim@juice-sh.op
4. Click Submit

Step 2: Analyze Security Question

You'll be asked a security question:

For jim@juice-sh.op, the question might be:

- "Your eldest siblings middle name?"
- "Mother's maiden name?"
- "Name of your favorite pet?"

Step 3: Research the Answer

OSINT (Open Source Intelligence):

For Juice Shop, security questions are intentionally weak. Try:

Common answers:

- For "favorite pet": [Zaya](#), [Max](#), [Buddy](#)
- For "sibling's name": [Samuel](#), [Brother](#)
- For Jim specifically: Check the Juice Shop GitHub for hints

The actual answer for jim@juice-sh.op: [Samuel](#)

Step 4: Reset Password

1. Enter the security answer: [Samuel](#)
2. Set a new password: [newpassword123](#)
3. Click Change

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Step 5: Login with New Password

1. Go back to login page
2. Email: `jim@juice-sh.op`
3. Password: `newpassword123`
4. Click Log in

You're now logged in as Jim!

Step 6: Verify Challenge Completion

1. Go to Account → Score Board
2. Find challenge: "Reset Jim's Password"
3. Should show  Solved

Step 7: Capture the Flag

Challenge Name: `Reset Jim's Password`

Flag Format: `NEXUS{RESET_JIM_PASSWORD_<YOUR_TEAM_ID>}`

Example:

`NEXUS{RESET_JIM_PASSWORD_TEAM05}`

Verification Checklist

- Successfully guessed Jim's security question answer
- Reset Jim's password
- Logged in as Jim with new password
- "Reset Jim's Password" challenge marked as Solved
- Submitted flag to Nexus

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Flag Submission Summary

All Flags for RED-1

IMPORTANT: Replace `<YOUR_TEAM_ID>` with your actual team ID (e.g., TEAM05)

Challenge	Flag Format	Points
SQL Injection - Admin Login	<code>NEXUS{SQLI_ADMIN_<YOUR_TEAM_ID>}</code>	20 XP
SQL Injection - Data Extraction	<code>NEXUS{SQLI_UNION_<USER_COUNT>_<YOUR_TEAM_ID>}</code>	25 XP
Reflected XSS	<code>NEXUS{XSS_REFLECTED_<YOUR_TEAM_ID>}</code>	20 XP
Stored XSS	<code>NEXUS{XSS_STORED_<YOUR_TEAM_ID>}</code>	20 XP
Password Reset	<code>NEXUS{RESET_JIM_PASSWORD_<YOUR_TEAM_ID>}</code>	15 XP
Total		100 XP

Where to Submit Flags

1. Login to Nexus Platform: <https://nexus.gdgpup.com>
2. Navigate to: Cybersecurity → Red Team → RED-1: Web Exploitation
3. Or use Cybersec Subdomain: <https://cybersec.nexus.gdgpup.com/rooms/red1>
4. Enter each flag in the respective challenge input box
5. Click "Validate Flag"

Submission Rules

- Case-sensitive - Flags must match exactly
- No spaces - Remove any spaces
- Unlimited attempts - No penalties for wrong submissions
- Instant feedback - System validates immediately
- Progress saved - Completed challenges auto-save

Proof of Completion

Required Screenshots

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To verify completion, you must submit ONE screenshot showing:

1. Juice Shop Score Board with challenges marked as "Solved"
2. Browser URL showing juice-shop.herokuapp.com
3. Your logged-in username (top-right corner)

How to submit screenshot:

1. Complete all 5 challenges
 2. Go to Account → Score Board
 3. Take a screenshot showing solved challenges
 4. Upload to Nexus platform under "Room Completion Proof"
-

Troubleshooting

Common Issues

Issue 1: Juice Shop is down

- Solution: Use alternative instance: <https://demo.owasp-juice.shop>

Issue 2: SQL injection isn't working

- Solution: Make sure you're using single quotes (') not backticks
- Check: Copy-paste the exact payload provided

Issue 3: XSS alert not showing

- Solution: Check if browser blocks pop-ups
- Alternative: Use `console.log()` instead of `alert()`

Issue 4: Can't find security answer

- Hint: For Jim's security question, the answer is [Samuel](#)
- Hint: Check Juice Shop documentation or source code

Issue 5: Challenge not marked as solved

- Solution: Refresh the Score Board page
- Check: Make sure you completed all steps correctly

> Learning Resources

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Before Starting Challenges

Read these FIRST:

1. PortSwigger SQL Injection Tutorial
 - URL: <https://portswigger.net/web-security/sql-injection>
 - Time: 30 minutes
 - Focus: Authentication bypass, UNION attacks
2. PortSwigger XSS Tutorial
 - URL: <https://portswigger.net/web-security/cross-site-scripting>
 - Time: 20 minutes
 - Focus: Reflected vs Stored XSS
3. OWASP Juice Shop Official Guide
 - URL: <https://pwning.owasp-juice.shop/>
 - Time: 15 minutes
 - Focus: Understanding the platform

During Challenges

Reference these as needed:

- SQL Injection Cheat Sheet: <https://portswigger.net/web-security/sql-injection/cheat-sheet>
- XSS Filter Evasion: <https://owasp.org/www-community/xss-filter-evasion-cheatsheet>
- Juice Shop Hints: Available on Score Board (click challenge for hints)

> Next Steps

After Completing RED-1

Continue Red Team Path:

- RED-2: Advanced Application Exploits (Session 9)
 - Requires completion of RED-1
 - File upload vulnerabilities
 - SSRF attacks
 - IDOR exploitation

Other Platforms:

- TryHackMe OWASP Top 10: <https://tryhackme.com/room/owasptop10>
- HackTheBox Web Challenges: <https://www.hackthebox.com/>
- PortSwigger Academy Labs: <https://portswigger.net/web-security/all-labs>

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If flag validation fails:

1. Double-check flag format
2. Verify team ID is correct
3. Contact team leads if still failing

Congratulations! 🎉

Achievement Unlocked: 🏆 Web Hacker

Sparky XP Earned: 100 XP

Next Challenge: RED-2: Advanced Web Exploitation

BLUE-1 FULL

DEVELOPMENT

UI/UX - Jem, Shai, Lloyd, Zybert
GitHub w/Docker - Jem

Authentication - Vincent
Supabase - Manel

Table	Key Fields
Users	gdg_id, name, role (Red/Blue), points
Challenges	id, title, category (Pwn, Web, Blue-Team), points
Submissions	id, gdg_id, challenge_id, flag_attempt, is_correct
Rooms	id, room_type (Red/Blue), container_url

Red Rooms

1. Zybert
 - JM
2. Zybert
 - JM
3. Josh
 - JL
4. Josh
 - JL

Blue Rooms

1. Shai
 - Nat
2. Shai
 - Nat
3. Jem
 - Victor
4. Jem
 - Victor

User Dashboard - All Front end & Vincent Backend

Backend Development (The Logic Engine)

The Backend team is responsible for the "Trust" layer—ensuring flags are valid and teams are credited correctly.

1. Flag Validation Logic (Supabase Edge Functions)

Since flags follow a specific formula (e.g., `NEXUS{SQLI_ADMIN_TEAM05}`), the backend needs a validator that:

- **Regex Check:** Verifies the string starts with `NEXUS{` and ends with `}`.
- **Dynamic Matching:** It shouldn't just check a static string. It needs to check if the `<GDG_ID>` in the flag matches the `gdg_id` of the logged-in user.
- **XP Injection:** Upon successful validation, increment the team's `sparky_points` by the specific challenge value (20 XP, 25 XP, etc.).

2. Database Schema (Supabase)

You need a relational structure to track the "RED-1" progress:

- **challenges Table:** Stores the challenge names (e.g., "SQLI_ADMIN"), XP values, and the "Base Flag" template.
- **submissions Table:** Tracks every attempt (timestamp, user_id, flag submitted, success/fail).
- **user_progress Table:** A many-to-many table showing which teams have completed which rooms.

3. File Storage (Supabase Storage) - GDrive Link

- Create a bucket for **"Proof of Completion."** The backend must handle the logic for users to upload their Juice Shop Scoreboard screenshots.

4. Authentication (Back-End + Front-End Pod)

- **Supabase Auth:** Assign one Back-End dev to configure GitHub/Discord OAuth in Supabase.
- **Protected Routes:** One Front-End dev ensures that a user cannot see the "RED-1" flags or instructions unless they are logged in and have a `TEAM_ID` assigned in their metadata.

Frontend Development (The Command Center)

The Frontend team (which is your largest group) needs to focus on the "Immersion" and "Clarity" of the lab instructions you provided.

1. The Room Interface (RED-1 Page)

- **Instruction Rendering:** Use a Markdown parser to render the detailed instructions you provided (Prerequisites, Tools, Step-by-Step).
- **Live Scoreboard Component:** A sidebar showing the "Top Teams" currently in RED-1, using **Supabase Realtime** so it updates without refreshing.

2. The Submission Portal

- **Interactive Input:** Create a text field for each challenge (Challenge 1, Challenge 2, etc.).
- **Dynamic Feedback:** Use "Success" toasts (green) for correct flags and "Error" toasts (red) for incorrect ones.
- **Proof Upload:** An upload component specifically for the "Verification Checklist" screenshot.

3. User Dashboard

- **XP Progress Bar:** A visual representation of the 100 XP total.
- **"Nexus Profile":** A section where the user can clearly see their **GDG_ID** (to prevent the "Where do I get my ID?" issue mentioned in your instructions).
- **Rooms Solved**

UI/UX

4. Lab Workspace (The "RED-1" View)

- **Instruction Panel (Left/Center):**
 - * **Header:** Room Title, Points (100 XP), and Estimated Time.
 - **Prerequisites:** List of required skills (HTTP, Browsers, SQL).
 - **Objectives:** Bulleted list of learning goals (SQLi, XSS, etc.).
 - **Step-by-Step:** Accordion-style sections for each of the 5 challenges.
- **Submission Panel (Right):**
 - **Flag Inputs:** Five text inputs labeled per challenge (e.g., "Challenge 1: Admin Login").
 - **Validation Button:** "Validate Flag" button per input.
 - **Proof Upload:** Drag-and-drop zone for the Juice Shop Scoreboard screenshot.

5. GDG Dept. (The "Community Hub")

- **Section 1: Chapters & Meetups:**

- Cybersecurity team under GDG PUP
- SJ 1 with pic
- SJ 2 with pic
- SJ 3 with pic
- SJ 4, with pic
- Coming Soon: Red vs Blue
- **Section 2: How it Works:** A four-step flow: "Pick Red vs Blue" -> "Attend Workshops" -> "Complete Rooms" -> "Earn XPs".
- **CTA:** "Join the department" button (Community GC via Messenger).

USER FLOW

